

Ch 1-4 Potential Quiz and Module 1 Exam Answers

A subset of these questions will appear on the Chapter 1-4 Quiz.

Quizzes are 30 minutes long and have 20 points.

A (probably different) subset of these questions will also appear on the Module 1 Exam, with questions from the Ch 1-4 and Ch 5 Quiz lists together making up at least half the points on the Module 1 Exam (i.e. at least 25 out of 50 points)

You should not expect Quiz or Exam questions to appear in the same order as in this document, nor should you expect the options for the answers to appear in the same order as in this document.

The remaining 50% of Module 1 Exam questions will be rephrasing or recombination of topics listed either in this document, in the Chapter 5 Potential Quiz questions, or the Ask Your Neighbor questions from class.

Questions are organized below into “pools” of similar questions; for each question it is shown how many points each of those questions would be worth should the question appear on the exam.

Answers are included for non-numeric questions in this document, but you should not count on being able to memorize an answer key for the exam, and you will not be allowed access to notes during the exam. You are strongly encouraged to form study groups to prepare for the exam, and you are encouraged to ask Gwen if you don't understand any concepts that are in this study sheet.

Pool 1: Each worth 1 point

1. Using the Chapter 4 model from class with C on the vertical axis, the increase in female labor force participation rates in the 1960s-1990s could be represented by (*ceteris paribus*)
 - a. [an increase/a **decrease**/no change/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [an increase/a decrease/no change/an **ambiguous-change**] in utility.
2. Using the Chapter 4 model from class, if there is a decrease in profits (π), this will, in equilibrium, lead to (*ceteris paribus*)
 - a. [an increase/a decrease/**no change**/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [an increase/a **decrease**/no change/an ambiguous-change] in utility.
3. Using the Chapter 4 model from class with C on the vertical axis, an increase in average number of years spent in school would best be represented by (*ceteris paribus*)
 - a. [**an increase**/a decrease/no change/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [an increase/a decrease/no change/an **ambiguous-change**] in utility.
4. Using the Chapter 4 model from class, if there is a decrease in after-tax real wages, this will, in equilibrium, lead to (*ceteris paribus*)
 - a. [an increase/a **decrease**/no change/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [an increase/a **decrease**/no change/an ambiguous-change] in utility.
5. Using the Chapter 4 model from class with C on the vertical axis, if younger workers value consumption more than older workers do, then aging of the labor force would best be represented by (*ceteris paribus*)
 - a. [**an increase**/a decrease/no change/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [an increase/a decrease/no change/an **ambiguous-change**] in utility
6. Using the Chapter 4 model from class, if there is a decrease in lump-sum taxes (T), this will, in equilibrium, lead to (*ceteris paribus*)
 - a. [an increase/a decrease/**no change**/an ambiguous-change] in the MRS out of consumption into leisure and
 - b. [**an increase**/a decrease/no change/an ambiguous-change] in utility.

Pool 2: Each worth 1 point

1. A good is normal for a consumer if

- A) it is always consumed in a consistent quantity.
- B) its consumption rises when income rises.
- C) its consumption falls when income rises.
- D) some minimal level of the good must be consumed to assure the consumer's survival.

Answer: B

2. We assume leisure is a normal good. This implies that

- A) an increase in taxes increases utility.
- B) we can use a representative household model.
- C) preferences over consumption are well defined.
- D) an increase in the wage increases demand for leisure.

Answer: D

3. The fact that indifference curves are downward sloping

- A) is not true.
- B) follows from the fact that more is preferred to less.
- C) follows from the property that the consumer likes diversity in his or her consumption bundle.
- D) follows from the property that consumption and leisure are normal goods.

Answer: B

4. The fact that indifference curves are bowed in toward the origin

- A) is not true.
- B) follows from the fact that more is preferred to less.
- C) follows from the property that the consumer likes diversity in his or her consumption bundle.
- D) follows from the property that consumption and leisure are normal goods.

Answer: C

5. The property of diminishing marginal rate of substitution follows from the property that the indifference curve is

- A) downward sloping.
- B) upward sloping.
- C) bowed in toward the origin.
- D) bowed out from the origin.

Answer: C

6. As the quantity of labor increases, the marginal product of capital

- A) is constant.
- B) increases.
- C) decreases.
- D) may either increase or decrease.

Answer: B

7. The assumption that the marginal product of labor decreases as the labor input increases implies that

- A) output decreases as the labor input increases.
- B) the wage increases as the labor input increases.
- C) the production function is concave.
- D) the production function shifts upward.

Answer: C

8. An increase in total factor productivity could be the result of

- A) adverse weather.
- B) the addition of new machinery.
- C) new hires.
- D) the introduction of new manufacturing methods.

Answer: D

9. If the firm hires more labor, everything else held constant, then

- A) the marginal product of labor falls.
- B) output decreases.
- C) there is an increase in the marginal product of labor.
- D) total factor productivity falls.

Answer: A

10. An increase in total factor productivity

- A) changes neither the slope nor the position of the production function.
- B) changes the slope but not the position of the production function.
- C) changes the position but not the slope of the production function.
- D) changes both the slope and the position of the production function.

Answer: D

11. The profit-maximizing quantity of labor equates the marginal product of labor with
A) total factor productivity.
B) the marginal product of capital.
C) the real wage.
D) the average product of labor.

Answer: C

12. In the one-period model, the assumption that everyone would behave the same way under the same circumstance is reflected in
A. All of these things.
B. Real wage equaling MPN.
C. The use of a representative consumer.
D. The use of a general equilibrium framework.

Answer: C

13. A constant returns to scale production function means

- A. $F(xK, N) = F(K, xN)$
- B. $F(xK, xN) = xF(K, N)$
- C. $F(xK, xN) = F(K, N)$
- D. All of these

Answer: B

14. Assume government cuts payroll taxes (proportional income taxes) and it leads to an increase in tax revenue collections. We would know from this that

- A. Substitution effects dominate income effects
- B. Income effects dominate substitution effects
- C. Markets are efficient
- D. None of these things can be known with certainty from the information supplied

Answer: A

15. If a forecaster uses a log-linear utility function to make a long run (twenty years from now) labor market forecast, the forecaster is likely going to

- A. Over-estimate the change in labor supply in response to a rising real wage
- B. Under-estimate the change in labor supply in response to a rising real wage
- C. Over-estimate the change in labor demand in response to a rising real wage
- D. Under-estimate the change in labor demand in response to a rising real wage

Answer: A

16. If wages remain constant over time, but we observe people working less, what can we say about the $MRS_{l,c}$?
- A. The MRS must have increased
 - B. The MRS must have decreased
 - C. The MRS must have remained the same
 - D. We cannot be sure what happened to the MRS

Answer: A

Pool 3: Worth 1 point

1. Math fill in the blank: Note that the numbers in brackets will not necessarily be the ones shown here, and that exams will draw randomly from different combinations of numbers:

Assume the following Cobb-Douglas aggregate production function

$$GDP = zK^{[0.7]}N^{[0.3]}$$

Where K stands for the aggregate capital stock and N is aggregate employment.

Ceteris paribus, if employment (N) rises by [2] percent, what is the growth rate of GDP?

Answer, rounding to two decimal points, using the following format: An answer of 10% (0.10) would be entered as *10* and an answer of 10.10% (.1010) would be entered as *10.10*

Pool 4: Each worth 3 points

1. Using the language and theory from class, assume that real wages **decrease** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. *[an increase/a **decrease**/no change/an ambiguous-change]* in labor supply from the substitution effect and
- b. *[an **increase**/a decrease/no change/an ambiguous-change]* in labor supply from the income effect.

The model predicts

- c. *[an increase/a **decrease**/no change/an ambiguous-change]* in Consumption from the substitution effect and
- d. *[an increase/a **decrease**/no change/an ambiguous-change]* in Consumption from the income effect.

Overall, the model predicts

- e. *[an increase/a decrease/no change/an **ambiguous-change**]* in labor supply,
- f. *[an increase/a **decrease**/no change/an ambiguous-change]* in Consumption, and
- g. *[an increase/a **decrease**/no change/an ambiguous-change]* in utility.

2. Using the language and theory from class, assume that proportional income tax rates “ t ” **increase** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. [an increase/a **decrease**/no change/an ambiguous-change] in labor supply from the substitution effect and
- b. [**an increase**/a decrease/no change/an ambiguous-change] in labor supply from the income effect.

The model predicts

- c. [an increase/a **decrease**/no change/an ambiguous-change] in Consumption from the substitution effect and
- d. [an increase/a **decrease**/no change/an ambiguous-change] in Consumption from the income effect.

Overall, the model predicts

- e. [an increase/a decrease/no change/**an ambiguous-change**] in labor supply,
- f. [an increase/a **decrease**/no change/an ambiguous-change] in Consumption, and
- g. [an increase/a **decrease**/no change/an ambiguous-change] in utility.

3. Using the language and theory from class, assume that proportional income tax rates “ t ” **decrease** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. [**an increase**/a decrease/no change/an ambiguous-change] in labor supply from the substitution effect and
- b. [**an increase**/a **decrease**/no change/an ambiguous-change] in labor supply from the income effect.

The model predicts

- c. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption from the substitution effect and
- d. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption from the income effect.

Overall, the model predicts

- e. [**an increase**/a decrease/no change/an **ambiguous-change**] in labor supply,
- f. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption, and
- g. [**an increase**/a decrease/no change/an ambiguous-change] in utility.

4. Using the language and theory from class, assume that real wages **increase** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. [**an increase**/a decrease/no change/an ambiguous-change] in labor supply from the substitution effect and
- b. [**an increase**/a **decrease**/no change/an ambiguous-change] in labor supply from the income effect.

The model predicts

- c. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption from the substitution effect and
- d. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption from the income effect.

Overall, the model predicts

- e. [**an increase**/a decrease/no change/**an ambiguous-change**] in labor supply,
- f. [**an increase**/a decrease/no change/an ambiguous-change] in Consumption, and
- g. [**an increase**/a decrease/no change/an ambiguous-change] in utility.

Pool 5: Each worth 3 points

1. Using the language and theory from class, assume that profits (π) **decrease** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. *[an increase/a decrease/**no change**/an ambiguous-change]* in labor supply from the substitution effect and
- b. *[**an increase**/a decrease/no change/an ambiguous-change]* in labor supply from the income effect.

The model predicts

- c. *[an increase/a decrease/**no change**/an ambiguous-change]* in Consumption from the substitution effect and
- d. *[an increase/**a decrease**/no change/an ambiguous-change]* in Consumption from the income effect.

Overall, the model predicts

- e. *[**an increase**/a decrease/no change/an ambiguous-change]* in labor supply,
- f. *[an increase/**a decrease**/no change/an ambiguous-change]* in Consumption, and
- g. *[an increase/**a decrease**/no change/an ambiguous-change]* in utility.

2. Using the language and theory from class, assume that lump-sum taxes **increase** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. *[an increase/a decrease/**no change**/an ambiguous-change]* in labor supply from the substitution effect and
- b. *[**an increase**/a decrease/no change/an ambiguous-change]* in labor supply from the income effect.

The model predicts

- c. *[an increase/a decrease/**no change**/an ambiguous-change]* in Consumption from the substitution effect and
- d. *[an increase/**a decrease**/no change/an ambiguous-change]* in Consumption from the income effect.

Overall, the model predicts

- e. *[**an increase**/a decrease/no change/an ambiguous-change]* in labor supply,
- f. *[an increase/**a decrease**/no change/an ambiguous-change]* in Consumption, and
- g. *[an increase/**a decrease**/no change/an ambiguous-change]* in utility.

3. Using the language and theory from class, assume that lump sum taxes **decrease** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. *[an increase/a decrease/**no change**/an ambiguous-change]* in labor supply from the substitution effect and
- b. *[an increase/**a decrease**/no change/an ambiguous-change]* in labor supply from the income effect.

The model predicts

- c. *[an increase/a decrease/**no change**/an ambiguous-change]* in Consumption from the substitution effect and
- d. *[**an increase**/a decrease/no change/an ambiguous-change]* in Consumption from the income effect.

Overall, the model predicts

- e. *[an increase/**a decrease**/no change/an ambiguous-change]* in labor supply,
- f. *[**an increase**/a decrease/no change/an ambiguous-change]* in Consumption, and
- g. *[**an increase**/a decrease/no change/an ambiguous-change]* in utility.

4. Using the language and theory from class, assume that profits (π) **increase** for the representative household. Consider the impact on labor, consumption, and utility:

The model predicts

- a. *[an increase/a decrease/**no change**/an ambiguous-change]* in labor supply from the substitution effect and
- b. *[an increase/**a decrease**/no change/an ambiguous-change]* in labor supply from the income effect.

The model predicts

- c. *[an increase/a decrease/**no change**/an ambiguous-change]* in Consumption from the substitution effect and
- d. *[**an increase**/a decrease/no change/an ambiguous-change]* in Consumption from the income effect.

Overall, the model predicts

- e. *[an increase/**a decrease**/no change/an ambiguous-change]* in labor supply,
- f. *[**an increase**/a decrease/no change/an ambiguous-change]* in Consumption, and
- g. *[**an increase**/a decrease/no change/an ambiguous-change]* in utility.

Pool 6: Each worth 1.5 points

1. Consider the following "shock" to the CRS firm profit-maximization problem from class where $Y=zf(K,N)$:

Suppose the government supplies firms with an **employment tax** t per unit of labor input N such that each unit of labor costs the firm $w(1+t)$. According to the model of the firm from class, we would expect to see

- a. [an increase/a **decrease**/no change/an ambiguous-change] in employment,
- b. [an increase/a **decrease**/no change/an ambiguous-change] in output,
- c. and an [an **increase**/a decrease/no change/an ambiguous-change] in the marginal productivity of labor.

2. Consider the following "shock" to the CRS firm profit-maximization problem from class where $Y=zf(K,N)$:

Suppose the government imposes a **production tax** (say, to pay for environmental cleanup) such that the firm pays t units of consumption goods to the government for each unit of output it produces.

According to the model of the firm from class, we would expect to see

- a. [an increase/a **decrease**/no change/an ambiguous-change] in employment,
- b. [an increase/a **decrease**/no change/an ambiguous-change] in output,
- c. and an [an **increase**/a decrease/no change/an ambiguous-change] in the marginal productivity of labor.

3. Consider the following "shocks" to the CRS firm profit-maximization problem from class where $Y=zf(K,N)$:

Suppose the government imposes a **production subsidy** s such that the firm earns s units of consumption goods from the government for each unit of output it produces.

According to the model of the firm from class, we would expect to see

- a. [an **increase**/a decrease/no change/an ambiguous-change] in employment,
- b. [an **increase**/a decrease/no change/an ambiguous-change] in output,
- c. and [an increase/a **decrease**/no change/an ambiguous-change] in the marginal productivity of labor.

4. Consider the following "shock" to the CRS firm profit-maximization problem from class where $Y = zf(K, N)$:

Suppose the government supplies firms with an *employment subsidy* s per unit of labor input N , such that the firm earns s units for each unit of labor it hires.

According to the model of the firm from class, we would expect to see

- a. [***an increase***/a decrease/no change/an ambiguous-change] in employment,
- b. [***an increase***/a decrease/no change/an ambiguous-change] in output,
- c. and an [***an increase***/**a decrease**/no change/an ambiguous-change] in the marginal productivity of labor.